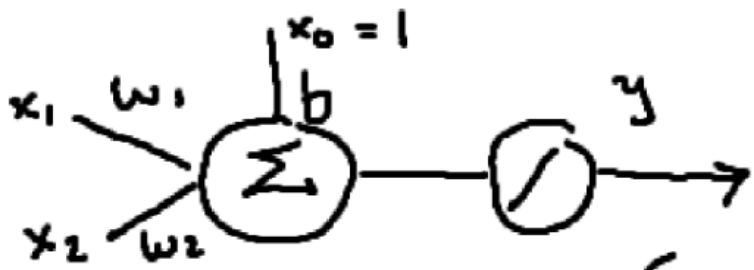
Question 1

Let's use a Perceptron with a linear activation function as below.



$$v = \sum_i x_i w_i + b \qquad y = \begin{cases} -1 & \text{if } v \leq 0 \\ 1 & \text{if } v > 0 \end{cases}$$

Set w1 = 1, w2 =1 and b=-1.

1. Write out the equation for the summation function v including the expanded version of the equation.
2. Complete the table below

x1	x2	V (activation function)	y(output)
-1	-1	-3	-1
-1	1	-1	-1
1	-1	-1	-1
1	1	1	1

3. Plot the input (x1,x2) against output y.
4. This network architecture functions as one of the LOGIC operators. Which one?

Question 2

Why do we need an activation function?

Question 1

$v = (x_1 w_1) + (x_2 w_2) + b$ $\quad w_1 = 1 \quad w_2 = 1 \quad b = -1$

$$y = \begin{cases} -1 & \text{if } v \le 0 \\ 1 & \text{if } v > 0 \end{cases}$$

$v = (x_1 (1)) + (x_2 (1)) + (-1)$

$v = x_1 + x_2 - 1$

i) $x_1 = -1 \quad x_2 = -1$

$v = -1 + (-1) - 1$
$= -3$

$y = -1$

ii) $x_1 = -1 \quad x_2 = 1$

$v = -1 + 1 - 1$
$= -1$

$y = -1$

iii) $x_1 = 1 \quad x_2 = -1$

$v = 1 + (-1) - 1$
$= -1$

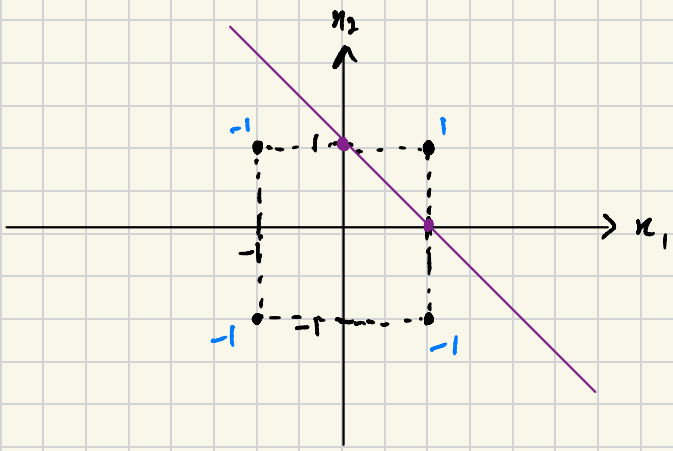
$y = -1$

iv) $x_1 = 1 \quad x_2 = 1$

$v = 1 + 1 - 1$
$= 1$

$y = 1$

Plot graph



$V = x_1 + x_2 - 1$

$v = 0$ (boundary)

$x_1 + x_2 - 1 = 0$

$x_2 = -x_1 + 1$

y-intercept = 1 $\quad$ x-intercept = 1

∴ This functions as an AND gate because it only output true state ($y=1$) when both x_1 and x_2 are true (1) while for other combination the output is false ($y=-1$).

Question 2

- We need activation function because it introduces non-linear properties, allowing neural network to learn complex patterns instead of as a single linear regression model.
- Activation function also is the threshold to bound the output.